Team Number: 104
Project Name: Slip Detection Subsystem
Team Member Names: Aakash Kumar
Version: 1.3

A. Major Components

All Major Components	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Microcontroller	PIC18F57Q43	1.8V - 5.5V	1	50	50	mA
	Accelerometer breakout	LIS3DH (SparkFun SEN-20659)	1.71V - 3.6V	1	2	2	mA

B. Power Rails

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Accelerometer breakout	LIS3DH (SparkFun SEN-20659)	1.71V - 3.6V	1	2	2	mA
	Subtotal					2.00	mA
	Safety Margin					0.25	
	Total Current Required on +3.3V Rail					2.50	mA
c1. Regulator or Source Choice	**+3.3V Regulator**	**LD1117V33**	**4.3V - 15V**	**1**	**800**	**800**	**mA**
	Total Remaining Current Available on +3.3V Rail					797.50	mA

+5V Power Rail	Component Name	Part Number	Supply	#	Absolute	Total	Unit
	Microcontroller	PIC18F57Q43	1.8V - 5.5V	1	50.00	50.00	mA
	Power LED (D1 + R1 1k)	Green LED	5V	1	3.00	3.00	mA
	Debug LEDs (D3, D4 + 220R)	Green / Red LED	5V (GPIO)	2	13.64	27.27	mA
	Push button pull-up (R2 10k)	10k resistor	5V	1	0.50	0.50	mA
	+3.3V Regulator Load	LD1117V33	4.3V - 15V	1	2.50	2.50	mA
	LD1117 quiescent current	LD1117V33	4.3V - 15V	1	10.00	10.00	mA
	Subtotal					93.27	mA
	Safety Margin					0.25	
	Total Current Required on +5V Rail					116.59	mA
c2. Regulator or Source Choice	**+5V Regulator**	**L7805**	**7V - 35V**	**1**	**1000**	**1000**	**mA**
	Total Remaining Current Available on +5V Rail					883.41	mA

C. Confirm Voltage Regulators

D. External Power Source

External Power Source 1	Component Name	Part Number	Supply	Out	Absolute	Total	Unit
Power Source 1 Selection	**9V Battery via Barrel Jack**	**4166-PART9VBATTER**	**+9V**	**+9V**	**500**	**500**	**mA**

Power Rails Connected to External Power Source 1						
+5V Regulator	**L7805**	**7V - 35V**	**116.59**	**116.59**	**mA**
L7805 quiescent current	**L7805**	**7V - 35V**	**8**	**8**	**mA**
Total Remaining Current Available on External Power Source 1				375.41	mA

E. Calculate Battery Life

Component Name	Part Number	Supply	Capacity	Required	
9V Battery via Barrel Jack	4166-PART9VBATTERY	+9V	500	124.59	
			Battery Life	4.01	hours